

Alejandro Solís Mercado

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SUMMARY

Mechanical engineer with a passion for technological innovation. I have experience in design, simulation, manufacturing, and programming. I specialize in developing functional prototypes using CAD, FEA/CFD simulation, and analytical optimization. I have led multidisciplinary teams, managed projects under pressure, and participated in engineering projects in international settings. I have participated in and led the development of RC aircraft for college competitions.

EDUCATION

Monterrey Institute of Technology, Toluca Campus

August 2022 – June 2026

Mechanical Engineering – Grade Average: 94/100 – Academic Excellence Scholarship: 70%

- **Key Courses:** Numerical modelling (MATLAB), CAD/CAM, Simulation FEM/CFD (ANSYS, Altair, AutoDesk, SolidWorks).

Universiteit Twente, Netherlands – Academic Exchange

August 2025 – January 2026

Minor in Aeronautical Management and Engineering

RELEVANT EXPERIENCE

Xinantec Aerodesign

August 2023 – July 2026

Interim President (2026)

Toluca, Mexico

- Demonstrated resilience and adaptation by assuming the interim presidency during a critical situation (4 weeks prior to the competition). Coordinated the design, manufacturing, and testing of a functional aircraft for Aerodesign MX 2026.

- 2nd best score in Design Report and Technical Presentation before judges among 17 international universities

Vice President and Lead Engineer (2023-2025)

- Developed engineering analysis using XFLR5, CFD, and MATLAB for the design optimization of a functional aircraft. Our team obtained 4th place nationally among 20 international universities in SAE Aerodesign MX 2023.

- Integrated aerodynamics, structures, and electronics teams for the full development of the system and the technical report.

- Coordinated sponsorships with Ansys, Altair, and Grupo SSC, and participated in STEM outreach initiatives.

DroneTeam Universiteit Twente

August 2025 – January 2026

Aerodynamics and Design Engineer

Enschede, Netherlands

- Collaborated in an international and multidisciplinary team, applying technical English to contribute to the development of an unmanned aerial vehicle for the UAS Challenge 2026, from conceptualization to a functional prototype.

- Designed aircraft components in CAD for 3D printing with lightweight materials (LW-PLA).

- Used SW Flow Simulation, MATLAB, and Flow5 for the aerodynamic optimization of the aerial vehicle.

General Motors

August 2024

Jr. Consultant – “Morphing Pocket” Challenge

Toluca, Mexico

- First place with the design and prototype of a smart map pocket for the automotive industry.

Borregos Toluca Baja Racing

April 2025

Mechanical Designer

Toluca, Mexico

- Designed and analytically validated the drivetrain shafts of a gearbox in Baja SAE using SolidWorks and MATLAB.

FRC Team High Altitude Robotics 9280

August 2022 – June 2024

Founding Mentor – Design and Programming

Toluca, Mexico

- Used and taught Java and C++ classes for autonomous routines, and SolidWorks for components and mechanisms.

- Contributed to obtaining the “Creativity Award” and “Rookie All-Star” awards in the team's first year.

FRC Team TecBot 3158

August 2018 – June 2022

Design and Programming Lead

Toluca, Mexico

- Developed mechanical components, FEA simulations, gear design, and control systems with Java.

- Contributed to industrial design awards by GM in 2019 and 2021, and autonomous code awards by Rockwell in 2019.

- Won the national “Dean’s List Finalist Award” in 2021 for my outstanding trajectory in leadership, commitment to competition values, and technical contribution. Only 6 Mexicans received this recognition.

CERTIFICATIONS, SKILLS, LANGUAGES, AND AWARDS

- **Certifications:** Certified SolidWorks Professional (CSWP), Scrum Fundamentals Certified.

- **Skills:** CAD Design (SolidWorks, Fusion360), Structural/Aerodynamic/Thermodynamic Simulation (FEM+CFD in ANSYS, Altair, AutoDesk, SW), Aeronautical Design (XFLR5, FLOW5), Programming (Java, Python, MATLAB, C++), Office.

- **Languages:** Spanish (Native), English (Advanced, C2), French (Basic), German (Basic).

- **Awards:** National Award for Excellence in Performance EGEL – CENEVAL (2026), Diploma of Excellence in Comprehensive Formation – ITESM (2026), Dean’s List Finalist Award – FRC Mexico (2021), 1st Place Statewide – Mexican Mathematics Olympiad (2021).